MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

THIRD SEMESTER

Discipline Specific Core Course (DSC)-MG3DSCCSC200- Database Management Systems

(2024 ADMISSION ONWARDS)

Duration : 1.5 Hours **Maximum Marks : 35**

1. Coding and Output - 20 Marks
2. Viva - 10 Marks
3. Record - 5 Marks

Q1. STUDENT TABLE

I. Create a table named STUDENT with the following structure:(5 Marks)

Column Name	Data Type	Constraints
enrollment_no	VARCHAR(10)	PRIMARY KEY
student_name	VARCHAR(50)	NOT NULL
department	VARCHAR(30)	NOT NULL
semester	INT	
cgpa	DECIMAL(3,2)	
city	VARCHAR(30)	

II. Insert the following records into the STUDENT table:(5 Marks)

enrollment_no	student_name	department	semseter	CGPA	city
CS001	Rahul Sharma	Computer Science	4	8.5	Delhi
CS002	Priya Singh	Computer Science	6	9.2	Mumbai
EC001	Amit Kumar	Electronics	4	7.8	Bangalore
ME001	Sneha Patel	Mechanical	2	8.0	Pune
CS003	Arjun Verma	Computer Science	4	6.5	Delhi
EC002	Divya Reddy	Electronics	6	9.5	Hyderabad
ME002	Rohan Gupta	Mechanical	4	7.2	Chennai
CS004	Kavya Nair	Computer Science	2	8.8	Kochi

III. Generate and execute the following queries

1. Add a new column phone_number of type VARCHAR(15) to the STUDENT table.(2 marks)
2. Update the CGPA of student 'Arjun Verma' (enrollment_no: CS003) to 7.0. (2marks)
3. Delete all students from the STUDENT table who have a CGPA less than 7.5. (2 marks)
4. Find the average CGPA of all students in the Computer Science department. (2 marks)
5. Display the number of students in each department. (2 marks)

Q2. Library Management System

I. Create a table BOOK with the following columns (5marks)

Column Name	Data Type	Constraints
book_id	VARCHAR(10)	PRIMARY KEY
title	VARCHAR(100)	NOT NULL
author	VARCHAR(50)	NOT NULL
genre	VARCHAR(30)	
price	DECIMAL(6,2)	
publisher	VARCHAR(50)	

II. Insert the following records to the table (5 Marks)

book_id	title	author	publisher	price
B001	Database Systems	Ramez Elmasri	Pearson	850.00
B002	Operating Systems	Abraham Silberschatz	Wiley	920.00
B003	Data Structures	Seymour Lipschutz	McGraw Hill	650.00
B004	Computer Networks	Andrew Tanenbaum	Pearson	780.00
B005	Python Programming	Mark Lutz	O'Reilly	1200.00
B006	Algorithms	Thomas Cormen	MIT Press	1500.00
B007	Web Development	Jon Duckett	Wiley	550.00
B008	Software Engineering	Roger Pressman	McGraw Hill	890.00

III. Generate and execute the following queries

1. Add a new column quantity to the table of type INT.(2 marks)
2. Modify the price of the book titled “Web Development” as 650.00.(2 marks)

3. Delete books with a price less than 300. (2 marks)
4. Find the average price of all books by publisher “Pearson”. (2 marks)
5. Display the details of the books in the order of price. (2 marks)

Q3.Employee Details

I. Create a table named EMPLOYEE with the following structure. (5 marks)

Column Name	Data Type	Constraints
emp_id	VARCHAR(10)	PRIMARY KEY
emp_name	VARCHAR(50)	NOT NULL
designation	VARCHAR(40)	
department	VARCHAR(30)	NOT NULL
salary	DECIMAL(10,2)	Should be greater than 0
joining_date	DATE	

II. Insert the following records into the EMPLOYEE table. (5 marks)

emp_id	emp_name	designation	department	salary	joining_date
E001	John Smith	Manager	Sales	65000.00	2020-03-15
E002	Sarah Johnson	Developer	IT	55000.00	2021-06-20
E003	Mike Brown	Analyst	Finance	48000.00	2019-08-10

E004	Emily Davis	Developer	IT	58000.00	2022-01-05
E005	David Wilson	Executive	Sales	42000.00	2021-11-12
E006	Lisa Anderson	Manager	Finance	70000.00	2018-05-25
E007	Tom Martinez	Developer	IT	52000.00	2023-02-18
E008	Anna Taylor	Executive	Sales	38000.00	2022-09-30

III. Generate and execute the following queries

1. Increase the size of department column to 35.(2 marks)
2. Update the salary of all employees in the IT department by increasing it by 10%.(2 marks)
3. Delete all employees from the EMPLOYEE table who have a salary less than 40000. (2 marks)
4. Find the maximum salary among all employees in the Sales department.(2 marks)
5. Display the average salary of employees in each department. (2 marks)

Q4. Inventory Management System

I. Create a table named PRODUCT with the following structure. (5marks)

Column Name	Data Type	Constraints
product_id	VARCHAR(10)	PRIMARY KEY
product_name	VARCHAR(60)	NOT NULL
category	VARCHAR(30)	NOT NULL
brand	VARCHAR(40)	
price	DECIMAL(10,2)	
stock_qty	INT	

II. Insert the following records into the PRODUCT table. (5 marks)

product_id	product_name	category	brand	price	stock_qty
P001	Laptop	Electronics	Dell	55000.00	25
P002	Smartphone	Electronics	Samsung	28000.00	40
P003	Washing Machine	Appliances	LG	32000.00	15
P004	Refrigerator	Appliances	Whirlpool	45000.00	12
P005	Headphones	Electronics	Sony	3500.00	60
P006	Microwave Oven	Appliances	Samsung	12000.00	20
P007	Tablet	Electronics	Apple	38000.00	18
P008	Air Conditioner	Appliances	Daikin	48000.00	8

III. Generate and execute the following queries

1. Add a new column warranty_period of type VARCHAR(20) to the PRODUCT table. (2 marks)
2. Update the price of all products in the Electronics category by reducing it by 5%.(2 marks)
3. Delete all products from the PRODUCT table where stock quantity is less than 10. (2 marks)
4. Display the products in Appliances category in the highest order of price.(2 marks)
5. Display the total stock quantity of products in each category. (2 marks)

Q5. Customer Order System

I. Create a table named ORDERS with the following structure. (5marks)

Column Name	Data Type	Constraints
order_id	VARCHAR(10)	PRIMARY KEY
customer_name	VARCHAR(50)	NOT NULL
product_name	VARCHAR(60)	NOT NULL
quantity	INT	
price_per_unit	DECIMAL(10,2)	

II. Insert the following data into the ORDERS table. (5marks)

order_id	customer_name	product_name	quantity	price_per_unit
O001	Rajesh Kumar	Laptop	2	52000.00
O002	Priya Sharma	Mobile Phone	1	25000.00
O003	Amit Patel	Tablet	3	18000.00
O004	Neha Singh	Headphones	5	2500.00